

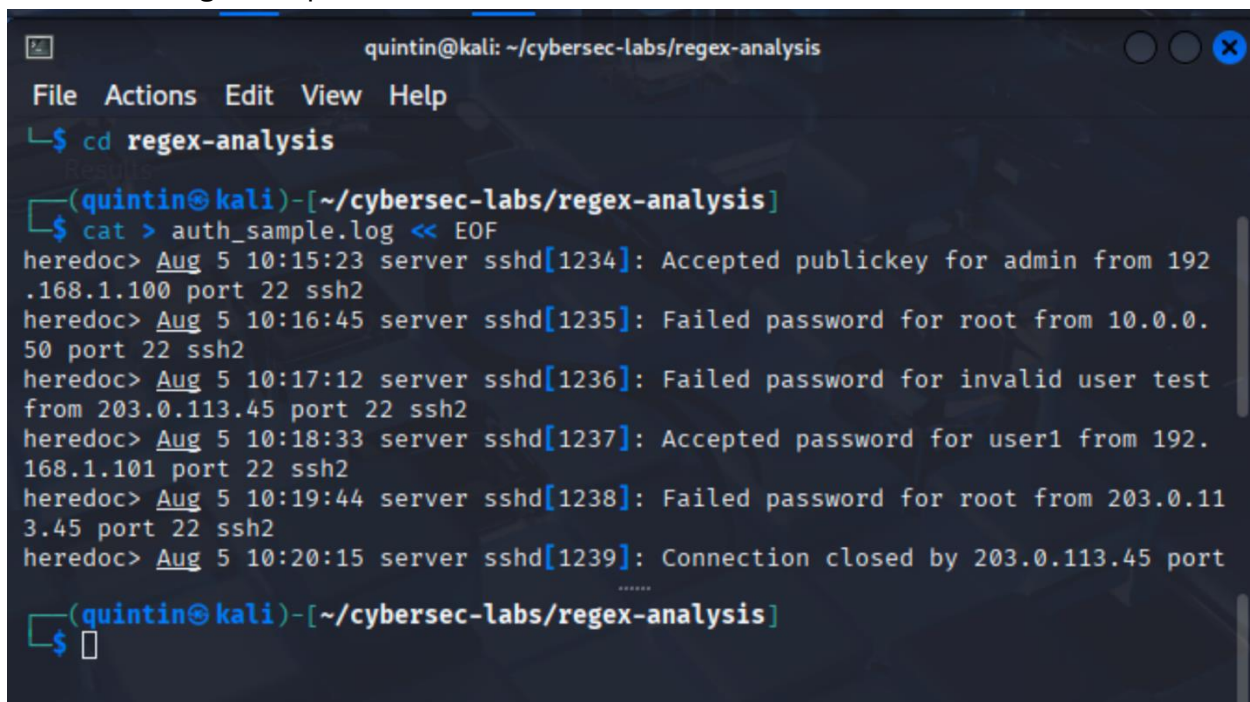
Lab 3: Linux Basics Part 3

Regular Expressions and Log Analysis

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Introduction: Regular expressions are an extremely valuable tool that cybersecurity professionals use for pattern matching in order to detect intrusion attempts, analyze malware behavior, and extract indicators of compromise (IoC) from log files. Regex expressions allow for quick identification of threats in large datasets where combing through an entire dataset might not be possible. Also, it's possible for penetration testers to use regex expressions to identify vulnerabilities within a system. This lab focuses on the use of regex applications in actual log analysis scenarios, centered around a given SOC's incident response plan.

Exercise 1: Regular Expression Fundamentals



```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
$ cd regex-analysis

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat > auth_sample.log << EOF
heredoc> Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100 port 22 ssh2
heredoc> Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22 ssh2
heredoc> Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user test from 203.0.113.45 port 22 ssh2
heredoc> Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101 port 22 ssh2
heredoc> Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port 22 ssh2
heredoc> Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port
*****
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$
```

```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "admin" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.10
0 port 22 ssh2
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (
using password: YES)

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "root" auth_sample.log
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 2
2 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 por
t 22 ssh2

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "Failed" auth_sample.log
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -i "failed" auth_sample.log
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -i "ERROR" auth_sample.log

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$
```

Exercise 2: Character Classes and Ranges

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9][0-9]:[0-9][0-9]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$
```



```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[A-Z]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "^[A-Z][a-z][a-z]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9][0-9]*\.[0-9][0-9]*\.[0-9][0-9]*\.[0-9][0-9]*" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
$ grep -o "[0-9][0-9]*\.[0-9][0-9]*\.[0-9][0-9]*\.[0-9][0-9]*" auth_sample.log
| sort | uniq >> ip_analysis.txt
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
```

```
$ cat ip_analysis.txt
== IP Address Analysis ==
```

All IP addresses found:

```
192.168.1.100
10.0.0.50
203.0.113.45
192.168.1.101
203.0.113.45
203.0.113.45
192.168.1.200
203.0.113.50
192.168.1.150
```

Unique IP addresses:

```
10.0.0.50
192.168.1.100
192.168.1.101
192.168.1.150
192.168.1.200
203.0.113.45
203.0.113.50
```


Exercise 3: Quantifiers and Repetition

```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
203.0.113.50

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "sshd\[([0-9]*)*\]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "port [0-9]\{2,5\}" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -o "sshd\[0-9\]{1,}\]" auth_sample.log
sshd[1234]
sshd[1235]
sshd[1236]
sshd[1237]
sshd[1238]
sshd[1239]
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -o "\[0-9\]{1,}\]" auth_sample.log
[1234]
[1235]
[1236]
[1237]
[1238]
[1239]
[5678]
[5679]
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9]{2}:[0-9]{2}" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

Exercise 4: Anchors and Position Matching


```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "[0-9][\{2\}:[0-9]\{2\}" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "^Aug" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "ssh2$" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
```



```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat test_file.txt
Valid line

Another valid line

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "^$" test_file.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -v "^$" test_file.txt
Valid line
Another valid line

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -c "^$" test_file.txt
2

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -cv "^$" test_file.txt
2
```

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "^[A-Z][a-z][a-z] *[0-9]" auth_sample.log
Aug 5 10:15:23 server sshd[1234]: Accepted publickey for admin from 192.168.1.100
port 22 ssh2
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22
ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user testfrom 203.0
.113.45 port 22 ssh2
Aug 5 10:18:33 server sshd[1237]: Accepted password for user1 from 192.168.1.101
port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port
22 ssh2
Aug 5 10:20:15 server sshd[1239]: Connection closed by 203.0.113.45 port 45123 [p
reauth]
Aug 5 10:21:30 server kernel: TCP: eth0: link up
Aug 5 10:22:10 server httpd[5678]: 192.168.1.200 - - [05/Aug/2024:10:22:10+0000]
"GET /admin HTTP/1.1" 404 152
Aug 5 10:23:25 server httpd[5679]: 203.0.113.50 - - [05/Aug/2024:10:23:25+0000] "
POST /login.php HTTP/1.1" 200 341
Aug 5 10:24:40 server mysqld: Access denied for user 'admin '@' 192.168.1.150' (u
sing password: YES)
```

```

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat failed_logins.txt
Failed Login Analysis
Failed SSH attempts:
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22 ssh2
Aug 5 10:17:12 server sshd[1236]: Failed password for invalid user test from 203.0.113.45 port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port 22 ssh2

Root login attempts:
Aug 5 10:16:45 server sshd[1235]: Failed password for root from 10.0.0.50 port 22 ssh2
Aug 5 10:19:44 server sshd[1238]: Failed password for root from 203.0.113.45 port 22 ssh2

```

Exercise 5: Advanced Pattern Matching

```

quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
Aug 5 10:30:16 server sshd[1241]: Failed password for admin from 198.51.100.10 port 22 ssh2
Aug 5 10:30:17 server sshd[1242]: Failed password for user from 198.51.100.10 port 22 ssh2
Aug 5 10:30:18 server sshd[1243]: Failed password for test from 198.51.100.10 port 22 ssh2
Aug 5 10:30:19 server sshd[1244]: Failed password for guest from 198.51.100.10 port 22 ssh2
Aug 5 10:35:22 server httpd[5680]: 203.0.113.100 - - [05/Aug/2024:10:35:22 +0000] "GET /admin/login.php HTTP/1.1" 200 1234
Aug 5 10:35:23 server httpd[5681]: 203.0.113.100 - - [05/Aug/2024:10:35:23 +0000] "POST /admin/login.php HTTP/1.1" 302 0
Aug 5 10:36:45 server httpd[5682]: 203.0.113.100 - - [05/Aug/2024:10:36:45 +0000] "GET /../../../../etc/passwd HTTP/1.1" 404 152
Aug 5 10:37:12 server httpd[5683]: 203.0.113.100 - - [05/Aug/2024:10:37:12 +0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87
Aug 5 10:40:30 server kernel: Firewall: IN=eth0 OUT= MAC=aa:bb:cc:dd:ee:ff SRC=203.0.113.200 DST=192.168.1.1 PROTO=TCP SPT=12345 DPT=1433
Aug 5 10:41:15 server suricata: [1:2100498:7] GPL SQL Injection attack [Classification: Web Application Attack] [Priority: 1]
Aug 5 10:42:00 server fail2ban: WARNING [sshd] Ban 198.51.100.10

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "Failed password.*from [0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}" security_events.log

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$

```



```

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "GET.*\\.\\.\\.\\.\\.\\.\\.\\.\\. security_events.log
+0000] "GET ../../../../etc/passwd HTTP/1.1" 404 152

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "UNION.*SELECT" security_events.log
+0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "GET.*['\"]" security_events.log
+0000] "GET /admin/login.php HTTP/1.1" 200 1234
+0000] "GET ../../../../etc/passwd HTTP/1.1" 404 152
+0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "DPT=[0-9]\\{4,5\\}" security_events.log
SRC=203.0.113.200 DST=192.168.1.1 PROTO=TCP SPT=12345 DPT=1433

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo "=== Security Threat Analysis ===" > threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo "SSH Brute Force Attempts:" >> threat_analysis.txt

```

```

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -c "Failed password.*from [0-9]\\{1,3\\}\\.[0-9]\\{1,3\\}\\.[0-9]\\{1,3\\}\\.[0-9]\\{1,3\\}" security_events.log >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo -e "\nSQL Injection Attempts:" >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "UNION.*SELECT\\|'.OR.*=\\|;.*DROP" security_events.log >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo -e "\nDirectory Traversal Attempts:" >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep "\\..\\.\\.\\.\\.\\.\\.\\.\\. security_events.log >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo -e "\nSuspicious IP Addresses:" >> threat_analysis.txt

```

```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
$ grep "\.\./\.\./\.\." security_events.log >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo -e "\nSuspicious IP Addresses:" >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -o "[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}\.[0-9]\{1,3\}" security_events.log | grep -E "^203\.0\.113\.|198\.51\.100\." | sort|uniq -c|sort -nr >> threat_analysis.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat threat_analysis.txt
== Security Threat Analysis ==
SSH Brute Force Attempts:
0

SQL Injection Attempts:
+0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87

Directory Traversal Attempts:
+0000] "GET ../../../../etc/passwd HTTP/1.1" 404 152

Suspicious IP Addresses:
  4 203.0.113.100
  1 203.0.113.200
```

Exercise 6: Extended Regular Expressions

```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "(Failed|Denied|Rejected)" security_events.log
Aug 5 10:30:15 server sshd[1240]: Failed password for root from
Aug 5 10:30:16 server sshd[1241]: Failed password for admin from
Aug 5 10:30:17 server sshd[1242]: Failed password for user from
Aug 5 10:30:18 server sshd[1243]: Failed password for test from
Aug 5 10:30:19 server sshd[1244]: Failed password for guest from

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "(admin|root|test|guest)" security_events.log
Aug 5 10:30:15 server sshd[1240]: Failed password for root from
Aug 5 10:30:16 server sshd[1241]: Failed password for admin from
Aug 5 10:30:18 server sshd[1243]: Failed password for test from
Aug 5 10:30:19 server sshd[1244]: Failed password for guest from
+0000] "GET /admin/login.php HTTP/1.1" 200 1234
+0000] "POST /admin/login.php HTTP/1.1" 302 0
```



```

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "UNION|SELECT|DROP|INSERT|DELETE" security_events.log
+0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "(\\.\\.\\.|etc|passwd|proc\\|sys\\/)" security_events.log
+0000] "GET /../..../etc/passwd HTTP/1.1" 404 152

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "DPT=(80|443|22|21|25|53|135|139|445|1433|3389)" security_events.log
SRC=203.0.113.200 DST=192.168.1.1 PROTO=TCP SPT=12345 DPT=1433

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "([0-9]{1,3}\\.){3}[0-9]{1,3}" security_events.log
198.51.100.10 port 22 ssh2
198.51.100.10 port 22 ssh2
198.51.100.10 port 22 ssh2
198.51.100.10 port 22 ssh2
198.51.100.10 port 22 ssh2
Aug 5 10:35:22 server httpd[5680]: 203.0.113.100 - - [05/Aug/2024:10:35:22
Aug 5 10:35:23 server httpd[5681]: 203.0.113.100 - - [05/Aug/2024:10:35:23
Aug 5 10:36:45 server httpd[5682]: 203.0.113.100 - - [05/Aug/2024:10:36:45
Aug 5 10:37:12 server httpd[5683]: 203.0.113.100 - - [05/Aug/2024:10:37:12
SRC=203.0.113.200 DST=192.168.1.1 PROTO=TCP SPT=12345 DPT=1433
Aug 5 10:42:00 server fail2ban: WARNING [sshd] Ban 198.51.100.10

```

```
quintin@kali: ~/cybersec-labs/regex-analysis
File Actions Edit View Help
y_events.log >> advance_threats.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ echo -e "\nNetwork scanning indicators:">> advance_threats.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ grep -E "DPT=(135|139|445|1433|3389|5985|5986)" security_events.log>>advance_threats.txt

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat advance_threats.txt
===== Advanced Threat Detection =====
High-risk authentication attempts:
Aug 5 10:30:15 server sshd[1240]: Failed password for root from
Aug 5 10:30:16 server sshd[1241]: Failed password for admin from
Aug 5 10:30:17 server sshd[1242]: Failed password for user from
Aug 5 10:30:18 server sshd[1243]: Failed password for test from
Aug 5 10:30:19 server sshd[1244]: Failed password for guest from

Web application attacks:
+0000] "GET ../../../../etc/passwd HTTP/1.1" 404 152
+0000] "GET /index.php?id=1' UNION SELECT * FROM users-- HTTP/1.1" 500 87

Network scanning indicators:
SRC=203.0.113.200 DST=192.168.1.1 PROTO=TCP SPT=12345 DPT=1433

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$
```

Exercise 7: Real-world Log Analysis Scenarios


```
(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat incident_analysis.txt
== Incident Response Analysis ==
Timeline of suspicious activities:
2024-08-05 10:45:12 firewall: BLOCK TCP 203.0.113.50:12345 → 192.168.1.150
2024-08-05 10:45:13 firewall: BLOCK TCP 203.0.113.50:12346 → 192.168.1.150
2024-08-05 10:45:14 firewall: BLOCK TCP 203.0.113.50:12347 → 192.168.1.150
2024-08-05 10:45:15 ids: ALERT [**] [1:2100498:7] SQL Injection attack [**]
2024-08-05 10:46:22 webserver: 203.0.113.75 - - [05/Aug/2024:10:46:22 +0000]
"GET /admin/config.php?id=1' OR '1'='1 HTTP/1.1" 500 1234

Network scanning attempts:

Web application attacks:
2024-08-05 10:45:15 ids: ALERT [**] [1:2100498:7] SQL Injection attack [**]
"GET /admin/config.php?id=1' OR '1'='1 HTTP/1.1" 500 1234

Malware indicators:
2024-08-05 10:48:15 malware_scanner: THREAT DETECTED: Trojan.Win32.Generic
2024-08-05 10:48:16 malware_scanner: QUARANTINE: /tmp/suspicious_file.exe
2024-08-05 10:49:00 dns: Query for suspicious-domain.com from 192.168.1.150
2024-08-05 10:49:01 dns: Blocked request to malware-c2.evil.com from 192.168.1.150

Attacker IP addresses:
2 Comp 3 203.0.113.75/sist
3 203.0.113.50
3 203.0.113.100

(quintin@kali)-[~/cybersec-labs/regex-analysis]
$ cat ioc_report.txt
== IoC Extraction ==
Malicious IP addresses:
203.0.113.100
203.0.113.50
203.0.113.75

Malicious domains:
suspicious-domain.com
malware-c2.evil.com
phishing-site.com
\Malicious files:
/tmp/suspicious_file.exe
/tmp/suspicious_file.exe
//phishing-site.com
2 Complete incident analysis:
```

Knowledge Assessment:

1. B
2. True
3. B
4. D
5. D

Conclusion:

This lab gave me valuable experience using regular expressions to analyze and detect security threats in log files. I learned how to build and apply different types of regex patterns including character classes, quantifiers, anchors, and extended expressions to extract specific data from large datasets. By combining these patterns with “grep”, I was able to identify failed login attempts, detect brute force and SQL injection patterns, and flag other suspicious activity hidden in system logs. These exercises helped me see how regex is used in real world cybersecurity tasks, like incident response, threat hunting, and log forensics and helped me practice creating automated analysis outputs, such as threat and IoC reports, that would be essential in a Security Operations Center (SOC) environment.